PERFORMANCE TESTING PHASE
Electricity Consumption Analysis

❖ 1.1Accuracy Testing

Accuracy testing was conducted to verify that the SQL query results, Tableau visualizations, and dashboard values accurately represented the electricity consumption dataset stored in the MySQL database.

The results of state-wise electricity usage, region-wise consumption, monthly trends, and top electricity-consuming states were cross-verified with the original dataset. All dashboard visualizations matched the SQL query outputs and the source dataset without discrepancies.

The testing confirmed that the dashboard provides reliable and accurate insights for electricity consumption analysis.

❖ 1.2 Performance Testing

Performance testing was performed to evaluate the responsiveness of the MySQL database, Tableau dashboard, and Flask web application.

The system was tested using the complete electricity consumption dataset containing **16,599 records**.

The results showed that:

- SQL queries executed efficiently.
- Tableau dashboards loaded quickly.
- Dashboard filters and interactions responded without noticeable delay.
- The Flask application successfully displayed the Tableau Public dashboard.
- The overall system performed efficiently under normal operating conditions.

The testing confirmed that the project meets the expected performance requirements.

❖ 1.3 Usability Testing

Usability testing was carried out to ensure that users could easily understand and interact with the dashboard.

The dashboard was evaluated for:

- Simple navigation
- Interactive filters
- Readable charts
- Proper labelling
- Clear dashboard layout
- Easy interpretation of insights

Users were able to identify electricity consumption trends, compare states and regions, and understand monthly usage without requiring advanced technical knowledge. The dashboard provided an intuitive and user-friendly experience.

❖ 1.4 Compatibility Testing

Compatibility testing was performed to verify that the project functions correctly across different software environments and platforms.

The project was successfully tested on:

- Windows Operating System
- Google Chrome Browser
- Microsoft Edge Browser
- MySQL Workbench
- Tableau Desktop
- Tableau Public
- Visual Studio Code
- Flask Local Server (http://127.0.0.1:5000)

The project worked successfully on all tested platforms without compatibility issues.

Performance Testing Summary Table

Testing Type	**Objective**	**Result**
Accuracy Testing	Verify correctness of SQL queries, dashboard values, and calculations	**Passed**
Performance Testing	Evaluate SQL execution, dashboard loading, and Flask response	**Passed**
Usability Testing	Ensure dashboard is easy to understand and interact with	**Passed**
Compatibility Testing	Verify functionality across software platforms and browsers	**Passed**

Test Cases

Test Case	**Expected Result**	**Actual Result**	**Status**
Import CSV into MySQL	Data imported successfully	16,599 records imported	✓ Passed
Execute SQL Queries	Correct analytical results	All queries executed successfully	✓ Passed
Connect Tableau with MySQL	Successful database connection	Connected successfully	✓ Passed
Create Dashboard	Interactive dashboard displayed	Dashboard created successfully	✓ Passed
Publish to Tableau Public	Dashboard published online	Published successfully	✓ Passed
Flask Integration	Dashboard displayed in browser	Displayed successfully	✓ Passed
GitHub Repository	Project files uploaded	Repository created successfully	✓ Passed

Overall Testing Result

The Electricity Consumption Analysis project was successfully tested for accuracy, performance, usability, and compatibility. All SQL queries executed correctly, Tableau visualizations accurately represented the dataset, and the dashboard was successfully published to Tableau Public. The Flask web application displayed the embedded dashboard without issues, and all planned functionalities worked as expected.